LANDPORT INTERNSHIP PROJECT BRIEF

Vehicle & Plate Detection for Landport Checkpoints

A Simple Computer Vision Prototype for Basic Gate Automation

Project Overview: This project is a simple, working Computer Vision prototype built to demonstrate how basic image processing and object detection can help landport checkposts keep track of incoming vehicles and read license plate numbers automatically.

1. Problem Statement

At landport entry gates, staff members often have to manually look at each vehicle, type out license plate numbers, and record entry details by hand. This process has a few simple problems:

- **Time-Consuming:** Typing plate numbers manually takes time and creates minor traffic delays at the entry gate.
- **Typing Errors:** Manual data entry can lead to small spelling or numerical mistakes in vehicle records.
- **Repetitive Work:** Staff spend a lot of time doing basic manual recording instead of focusing on general security and management.

2. Proposed Solution

To solve this, I built a simple Python script using standard Computer Vision libraries (OpenCV and EasyOCR / Haar Cascades):

- **Detect Incoming Vehicles:** Uses a camera feed or image/video input to identify when a vehicle is at the gate.
- **Read License Plates (ANPR):** Automatically crops the number plate region and extracts the text using Optical Character Recognition (OCR).
- **Log Records Automatically:** Saves the extracted number plate text with a time stamp into a basic database or CSV file.

3. Problem vs. Solution Summary

Manual Checkpoint Process	Simple Computer Vision Solution
Officer manually writes or types number plates	OCR automatically extracts plate text from camera image
Prone to human typing errors	Direct digital text logging eliminates manual typos
Manual entry takes ~1 to 2 minutes per vehicle	Automated detection and OCR takes just a few seconds

Submitted for Landport Internship Evaluation • Problem Statement & Solution Document